

Jordan Mark **Barbone, M.A.**

DATA SCIENTIST, R DEVELOPER

West Chester, PA

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R programmer with experience in statistical programming, monitoring, and report automation in cognitive and CNS clinical trials

Skills

R - Advanced	package development, data analysis, statistical programming, data visualization, ETL/Data integration, tidyverse (dplyr, ggplot2, purrr, tidyr, devtools, etc), Rmarkdown, quarto
Other programming experience	Intermediate: Python (pyspark, pandas) Basic: SQL, git, bash
Clinical trials	clinical data management programming, centralized monitoring, eCOA, data review, Rater training and qualification
Psychology	behavioral statistics, research methods, cognitive psychology, learning psychology

Experience

Data Scientist

Jun 2021 - Present

COGSTATE

New Haven, CT (Remote)

- Independently developed internal **R packages** to advance programming, visualization, normative data management, central monitoring, data access, and general utilities
- Designed and development **R shiny** applications for displaying central monitoring activity and performing interactive analyses
- Refactored Data Engineering **PySpark** code into a **python** package for data transformations within **Azure Databricks** and **DeltaTable** management
- Refactored legacy **SQL** code to seamlessly integrate **PySpark** functionalities and enforce best coding practices
- Designed and implemented an efficient process for managing and updating and tracking normative data, collaborating closely with Science and Product Management teams
- Authored comprehensive standards and protocols for **Jira** workflows and **git** management, ensuring consistency and efficiency
- Developed and managed a robust daily and weekly reporting system with **R tidyverse** scripts for the monitoring of a high-profile international Alzheimer's Disease trial
- Analyzed and interpreted requirements from Science and Data Management to drive strategic advancements within our Data Lakehouse
- Developed, reported, and presented analyses with **R tidyverse** scripts for custom data queries related to both ongoing and concluded clinical, providing actionable insights for rater training and central monitoring
- Developed a **python** data extraction, reformatting, and storage processes for academic research studies, greatly enhancing accessibility for end-users
- Contributed to the creation of poster materials for industry conferences and presentations, showcasing research, developments, and services

Principal Data Analyst

Dec 2019 - Jun 2021

MADRIGAL PHARMACEUTICALS

Conshohocken, PA

- Developed internal **R packages** with **tidyverse** features for the creation of daily metric and reports
- Aggregated and organized data sources in STDM, ADaM, TFLS, and pre-finalized formats; compiled source data from multiple vendor data bases and reports; reconciled differences between databases using **R tidyverse** suite
- Analyzed efficacy, safety, and exploratory data for: ongoing phase 3 studies in NASH and NAFLD; ad hoc analyses for phase 2; biomarker and patient reported outcomes with **R**
- Reviewed and reported on statistical testing in **R** such as: ANCOVA, LS-Means, ROC, Kruskal-Wallis, Wilcoxon rank sum, t-test, correlations (Pearson product, Spearman rho, Kendall's tau)
- Trained and onboarded data analyst

Clinical Data Scientist

Jan 2019 - Nov 2019

SIGNANT HEALTH (FORMERLY BRACKET AND CRF HEALTH)

Wayne, PA

- Interacted with and analyzes CNS, psychiatry, and neurology clinical trial data quality and outcomes information for clinical insights
- Developed and implemented new data products and deliverables for blinded, statistical data monitoring and review; maintained and enhanced current data products; advised new analytic and machine learning initiatives; identified data sources and aggregation processes; defined and updated processes for deliverables; point of contact for analytic deliverables; presented information using data visualization techniques
- Reviewed and developed analyses with SME for detection of errors for study endpoints; collaborated with review of analytic findings
- Trained and supervised of Data Scientists; developed and provided training for new and current processes, programming, and technologies/software; lead, organized team meetings and status updates; wrote and reviewed new process documentation.
- Assisted Program Managers and Project Delivery Managers for process implementation for data products and deliverables for blinded, statistical data monitoring and review; provided consultation for programming and development for Clinical Data Management; liaised between Data Services and Clinical Science teams; provided analytic and research support for Clinical Science

Clinical Data Specialist

Jan 2018 - Dec 2018

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Interacted with and analyzes CNS, psychiatry, neurology clinical trial data quality and outcomes information for clinical insights; collaborated with other Bracket stakeholders and teams to execute and provide ongoing analysis related to study deliverables or internal needs
- Developed training material development and QC; performed data review, eCOA user acceptance testing; study staff contact, sponsor contact, contact with outside experts, report writing.
- Maintained, classified clinical information/materials; reviewed scale administration/scoring; provided clinical feedback, remediation for raters

Clinical Associate

May 2016 - Dec 2017

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Participated in applicable aspects of clinical content development and delivery, including training material development and QC, data review, eCOA user acceptance testing, study staff contact, sponsor contact, contact with outside experts, and report writing
- Maintained, classified clinical information/materials for CNS, psychiatry, neurology clinical trials
- Provided data review for Bracket's Endpoint Reliability programs; clinical analysis on study endpoint data for administration and scoring calibration to standard training principles; and provided remediation as needed to recalibrate to study standards.

Clinical Assistant

Aug 2015 - May 2016

SIGNANT HEALTH (BRACKET)

Wayne, PA

- Developed object specific video material, clinical content for training and qualification programs for study staff in clinical trials; addressed and reviewed data quality issues; clinical scheduling; ensured compliance with quality assurance, participate in reporting, analysis activities and research activities (e.g., collection and dissemination of literature and/or data); prepared reports of clinical activities.

Writing Tutor

Aug 2014 - May 2015

LEARNING ASSISTANCE & RESOURCE CENTER

West Chester University of Pennsylvania,

West Chester, PA

- Tutored in 1-2 person sessions to develop skills necessary for effective writing, as well promoted independent and active learning.
- Monitored efficiency of the Academic Development Program and growth of students through maintaining daily session activity logs, utilizing Bloom's Taxonomy, biweekly student reports, and semester evaluations.

Education

Master of Arts

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2017 - May 2019

- General Psychology, Advisor: Geeta Shivde, Ph.D.
- Thesis: The effects of participant-selected background music on executive function task performance
- GPA: 4.00

Non-degree Graduate Student

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2016 - Jun 2017

- Classes in Statistics, Research Methods, and Psychometrics
- GPA: 4.00

Bachelor of Arts

West Chester, PA

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Aug 2011 - May 2015

- Major in Psychology, Minor in Biology
- Cumulative GPA: 3.470
- Major GPA: 3.834

Papers

PEER REVIEWED PUBLICATIONS

Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, & Harrison, S. A. (2021). Hepatic fat reduction due to resmetirom in patients with nonalcoholic steatohepatitis is associated with improvement of quality of life. *Clinical Gastroenterology and Hepatology*. <https://doi.org/10.1016/j.cgh.2021.07.039>

Solomon, T. M., **Barbone, J. M.**, Feaster, H. T., Miller, D. S., deBros, G. B., Murphy, C. A., & Michalczyk, D. (2019). Comparing the standard and electronic versions of the Alzheimer's disease assessment scale—cognitive subscale: A validation study. *The Journal of Prevention of Alzheimer's Disease*, 6(4), 237–241. <https://doi.org/10.14283/jpad.2019.27>

Irani, F., **Barbone, J. M.**, Beausoleil, J., & Gerald, L. (2017). Is asthma associated with cognitive impairments? A

meta-analytic review. *Journal of Clinical and Experimental Neuropsychology*, 39(10), 965–978. <https://doi.org/10.1080/13803395.2017.1288802>

MASTER'S THESIS

Barbone, J. M. (2019). *The effects of participant-selected background music on executive function task performance* [Master's thesis, West Chester University of Pennsylvania]. <https://doi.org/10.13140/RG.2.2.34109.41446>

Posters and presentations

COGSTATE - DIGITAL COGNITION

Tronchin, G., Kingery, L., Leventhal, R., Bartolic, E., Wacker, S., **Barbone, J. M.**, & Wessels, A. M. (2024, March). Central monitoring of rater performance and characteristics of efficacy and assessments in the TRAILBLAZER-ALZ 2 study. *Poster Presented at the International Conference on Alzheimer's and Parkinson's Disease (ADPD), Lisbon, Portugal, and Online.*

Barbone, J. M., Khurshid, K., Edgar, C. J., & Maruff, P. (2023, July). Equating the ADAS-Cog and MMSE to the Cogstate Brief Battery scores. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Amsterdam, Netherlands.* <https://doi.org/doi/10.1002/alz.077060>

Edgar, C. J., Maruff, P., Harrison, J. E., **Barbone, J. M.**, Leventhal, R., & Alam, J. j. (2022). Validity and reliability of a composite cognitive outcome measure for clinical trials in dementia with lewy bodies. *Poster Presented at the ADPD Conference, Barcelona, Spain.* Video: <https://youtu.be/N8TbGImGV5U>. <https://www.cogstate.com/wp-content/uploads/2022/04/00171-Validity-and-reliability-of-a-composite-cognitive-outcome-in-d273-Read-Only.pdf>

Wacker, S., & **Barbone, J. M.** (2022). Rater perspectives on Applied Training of cognitive clinical outcome assessments, delivered by Neuropsychology experts. *Poster Presented at AAIC, San Diego, California, USA.* <https://www.cogstate.com/wp-content/uploads/2022/08/Rater-Perspectives-on-Applied-Training-of-Cognitive.pdf>

Edgar, C. J., Amariglio, R., **Barbone, J. M.**, Chandler, J., Coons, S., Donohue, M., Lenderking, W., & Sperling, R. (2022, December). Preliminary evidence for reliability and validity of the Interpersonal Functioning and Daily Activities Questionnaire (IFDAQ) in the A4/LEARN pre-randomization sample. *Slides Presented at Clinical Trials on Alzheimer's Disease (CTAD) Conference, San Francisco, California, USA.*

MADRIGAL - LIVER DISEASE

Harrison, S. A., Taub, R., **Barbone, J. M.**, Franc, J., & Karsdal, M. A. (2020, March). Resmetirom, a beta selective thyroid hormone receptor agonist, reduces net collagen III deposition in nonalcoholic Steatohepatitis. *Poster to Have Been Presented at American Association for the Study of Liver Diseases (AASLD) Emerging Topic Conference 2020, Nuclear Receptors in Nonalcoholic Fatty Liver Diseases (Conference Cancelled).*

Harrison, S. A., Taub, R. A., Karsdal, M. A., Franc, J., Bashir, M., **Barbone, J. M.**, Neff, G., Gunn, N. T., & Moussa, S. (2020, November). Algorithm for predicting advanced NASH fibrosis on screening biopsy in resmetirom phase 3 MAESTRO-NASH clinical trial. *Poster Presented at the AASLD Liver Meeting Digit Experience Conference.*

Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, Moussa, S., & Harrison, S. A. (2020, November). Improvement of health-related quality of life is associated with improvement of fat fraction by MRI-PDFF in patients with nonalcoholic steatohepatitis treated with resmetirom. *Poster Presented at the AASLD Liver Meeting Digit Experience Conference.*

SIGNANT HEALTH (BRACKET) - ECOA DATA QUALITY

Crittenden, K., Machizawa, S., Feaster, H. T., **Barbone, J. M.**, Hong, B., Verma, P., & Zhou, W. (2020, April). Cross-cultural differences in PANSS item ratings: Comparisons of six geo-cultural regions. *Poster Presented at the 2020 Schizophrenia International Research Society (SIRS) Conference, Florence, Italy.* <https://doi.org/10.1093/schbul/sbaa031.103>

Roy, M., Brown, J., Hong, B., Benecke, R., Chen-Tackett, Z., Zhou, W., **Barbone, J. M.**, Feaster, H. T., & Sachs, G. (2020,

- September). eCOA prompted MADRS interview: Balancing through assessment and efficiency. *Poster Presented at ISCTM*.
- Feaster, H. T., **Barbone, J. M.**, Miller, D. S., & Solomon, T. M. (2019, June). Rater remediation on the ADAS-cog leads to longitudinal improvement in clinical trial data quality. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.3923>
- Karas, S. M., **Barbone, J. M.**, Solomon, T. M., Hong, B., & Feaster, H. T. (2019, June). Impact of data quality programs on MMSE inclusion criteria in Alzheimer's disease clinical research trials. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.862>
- Solomon, T. M., Feaster, H. T., Karas, S. M., Garcia-Valdecasas, M., **Barbone, J. M.**, DiGregorio, D. T., DeBonis, D., & Miller, D. S. (2019, July). The evolution of technology in data quality programs for Alzheimer's disease clinical trials: What have we learned and where are we going. *Poster Presented at the Alzheimer's Association International Conference (AAIC), Los Angeles, CA, USA*. <https://doi.org/10.1016/j.jalz.2019.06.4332>
- Seichepine, D. R., Solomon, T. M., Jacobs, E. B., **Barbone, J. M.**, DiGregorio, D. T., & Miller, D. S. (2018, February). Comparison between "flat" and "enhanced" eCOA of MMSE Attention and Calculation in clinical trials of Alzheimer's disease. *Presented at the ISCTM 14th Annual Scientific Meeting, Washington, D.C., USA*. <https://doi.org/10.13140/RG.2.2.16193.61289>
- Solomon, T. M., **Barbone, J. M.**, Miller, D. S., & Feaster, H. T. (2018, June). Pilot validation of an Electronic Alzheimer's Disease Assessment Scale - Cognitive Subscale (eADAS-Cog). *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.719>
- Feaster, H. T., **Barbone, J. M.**, Garcia-Valdecasas-Colell, M., & Solomon, T. M. (2018, July). Electronic ADAS-Cog (eADAS-Cog) data quality: How do countries compare? *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.1373>
- Karas, S. M., **Barbone, J. M.**, DeBonis, D., & Solomon, T. M. (2018, July). The use of statistical modeling to complement data quality programs. *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.066>
- Solomon, T. M., Karas, S. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., & Feaster, H. T. (2018, July). Analysis of the rates and types of errors on the enhanced eCOA version of the Alzheimer's Disease Assessment Scale - Cognitive subscale and Mini-mental State Examination used in dementia clinical trials. *Poster Presented at the 2018 Alzheimer's Association International Conference, Chicago, IL, USA*. <https://doi.org/10.1016/j.jalz.2018.06.070>
- Barbone, J. M.**, Solomon, T. M., Feaster, H. T., Garcia-Valdecasas-Colell, M., & Miller, D. S. (2018, October). MMSE screening data quality for Alzheimer's disease studies across countries. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain*. <https://doi.org/10.13140/RG.2.2.27921.68966>
- Garcia-Valdecasas-Colell, M., **Barbone, J. M.**, Solomon, T. M., Feaster, H. T., & Tott, N. (2018, October). Identifying impact of rater change on MMSE and CDR data in multi-national Alzheimer's disease clinical trials. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain*. <https://doi.org/10.13140/RG.2.2.21210.80327>
- Wessels, A. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., Mullen, J. A., Sims, J. R., & Solomon, T. M. (2018, October). Lanabecestat- Rater performance and error characteristics of efficacy assessments in the DAYBREAK-ALZ study. *Presented at the 11th Clinical Trials on Alzheimer's Disease (CTAD), Barcelona, Spain*. <https://bit.ly/32p18au>
- Butler, A., Feaster, H. T., Miller, D. S., & **Barbone, J. M.** (2017, May). Using iterative user experience design to improve electronic clinical outcomes assessment data quality. *Presented at the ISPOR 22nd Annual International Meeting*,

Boston, MA, USA. <https://bit.ly/3isANeP>

Feaster, H. T., Solomon, T. M., Abi-Saab, D., Vogt, A., **Barbone, J. M.**, Harrison, J., & Miller, D. S. (2017, July). The impact of electronic clinical outcome assessments (eCOA) on Alzheimer's disease clinical trial data quality. *Poster Presented at the Alzheimer's Association International Conference, London, England, UK.* <https://doi.org/10.1016/j.jalz.2017.06.1828>

Enger, B., & **Barbone, J. M.** (2017, October). Distribution of performance and incorrect ratings in qualification video scoring. *Poster Presented at CNS Summit, Boca Raton, FL, USA.* <https://doi.org/10.13140/RG.2.2.14096.46088>

Solomon, T. M., **Barbone, J. M.**, Karas, S. M., & Feaster, H. T. (2017, October). Longitudinal impact of audio review on data quality. *Poster Presented at Clinical Trials on Alzheimer's Disease Conference (CTAD), Boston, MA, USA.* <https://bit.ly/2DYaFtd>

Solomon, T. M., Feaster, H. T., **Barbone, J. M.**, & Miller, D. S. (2017, November). Utilizing audio review to improve ADCS-ADL data quality. *Poster Presented at Clinical Trials on Alzheimer's Disease Conference (CTAD), Boston, MA, USA.* <https://bit.ly/3m1Ku0c>

WCUPA - ASTHMA AND MUSIC COGNITIVE

Barbone, J. M., & Shivde, G. (2019, May). The effects of participant selected background music on executive task performance. *Poster Presented at the Association for Psychological Science, Washington, D.C., USA.* <https://doi.org/10.13140/RG.2.2.26125.79842>

Barbone, J. M. (2018, November). Trends in self-reported use of music while studying: Implications for research. *Poster Presented at the Research and Creative Activity Day at West Chester University, West Chester, PA, USA.* <https://doi.org/10.13140/RG.2.2.14075.57124>

Barbone, J. M., Saxena, S., & Irani, F. (2015, February). Neuropsychological performance in individuals with asthma: Two meta-analyses. *Poster Presented at the Eastern Psychological Association Conference, Philadelphia, PA, USA.* <https://bit.ly/3hu6J0i>

Mulligan, R. D., **Barbone, J. M.**, Saxena, S., Carey-Jr, D., Clappsy, J., & Irani, F. (2015, March). Asthma cognition data collection. *Poster Presented at the West Chester University Psychology Research Day, West Chester, PA, USA.* <https://bit.ly/2ZAD5kn>

Barbone, J. M., Saxena, S., & Irani, F. (2015, July). Does asthma effect neuropsychological performance? A meta-analysis. *Poster Presented at the American Psychological Association Convention, Toronto, Canada.*

Irani, F., **Barbone, J. M.**, & Beausoleil, J. (2015, August). The impact of asthma on cognitive functioning. *Paper Presented at the American Psychological Association Convention, Toronto, Canada.*

CRAN packages

Barbone, J. M. (2024). *Fuj: Functions and utilities for jordan.* <https://jmbarbonate.github.io/fuj/>

Barbone, J. M. (2024). *Mark: Miscellaneous, analytic r kernels.* <https://github.com/jmbarbonate/mark>

Barbone, J. M., & Garbuszus, J. M. (2024). *openxlsx2: Read, write and edit xlsx files.* <https://janmarvin.github.io/openxlsx2/>

Barbone, J. M. (2023). *Echo: Echo code evaluations.* <https://github.com/jmbarbonate/echo>

Barbone, J. M. (2023). *Scribe: Command argument parsing.* <https://jmbarbonate.github.io/scribe/>